

Michael Baron

PROBABILITY AND STATISTICS FOR COMPUTER SCIENTISTS

SECOND EDITION



$$\lambda_A \pi_{i-1} = \lambda_S \pi_i$$

239

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Michael Baron

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CRC Press

Taylor & Francis Group

Boca Raton London New York

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Taylor & Francis Group, an **informa** business

A CHAPMAN & HALL BOOK

CRC Press
Taylor & Francis Group
6000 Broken Sound Parkway NW, Suite 300
Boca Raton, FL 33487-2742

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Version Date: 20130625

International Standard Book Number-13: 978-1-4822-1410-9 (eBook - ePub)

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To my parents – Genrietta and Izrael-Vulf Baron

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Chapter 4

Continuous Distributions

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Recall that any discrete distribution is concentrated on a finite or countable number of isolated values. Conversely, *continuous variables can take any value of an interval*, (a, b) , $(a, +\infty)$, $(-\infty, +\infty)$, etc. Various times like service time, installation time, download time, failure time, and also physical measurements like weight, height, distance, velocity, temperature, and connection speed are examples of continuous random variables.

4.1 Probability density

For all continuous variables, the probability mass function (pmf) is always equal to zero,¹

$$P(x) = 0 \quad \text{for all } x.$$

As a result, the pmf does not carry any information about a random variable. Rather, we can use the *cumulative distribution function* (cdf) $F(x)$. In the continuous case, it equals

$$F(x) = \mathbf{P}\{X \leq x\} = \mathbf{P}\{X < x\}.$$

These two expressions for $F(x)$ differ by $P\{X = x\} = P(x) = 0$.

In both continuous and discrete cases, the cdf $F(x)$ is a non-decreasing function that ranges from 0 to 1. Recall from Chapter 3 that in the discrete case, the graph of $F(x)$ has *jumps* of magnitude $P(x)$. For continuous distributions, $P(x) = 0$, which means no jumps. The cdf in this case is a continuous function (see, for example, Fig. 4.2 on p. 77).

¹**Remark:** In fact, any probability mass function $P(x)$ can give positive probabilities to a finite or countable set only. The key is that we must have $\sum_x P(x) = 1$. So, there can be at most 2 values of x with $P(x) \geq 1/2$, at most 4 values with $P(x) \geq 1/4$, etc. Continuing this way, we can list all x where $P(x) > 0$. And since all such values with positive probabilities can be listed, their set is at most countable. It cannot be an interval because any interval is uncountable. This agrees with $P(x) = 0$ for continuous random variables that take entire intervals of values.

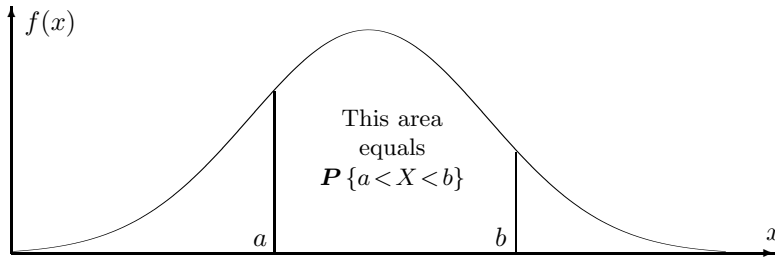


FIGURE 4.1: Probabilities are areas under the density curve.

Assume, additionally, that $F(x)$ has a derivative. This is the case for all commonly used continuous distributions, but in general, it is not guaranteed by continuity and monotonicity (the famous Cantor function is a counterexample).

DEFINITION 4.1

Probability density function (pdf, density) is the derivative of the cdf, $f(x) = F'(x)$. The distribution is called **continuous** if it has a density.

Then, $F(x)$ is an *antiderivative* of a density. By the Fundamental Theorem of Calculus, the integral of a density from a to b equals to the difference of antiderivatives, i.e.,

$$\int_a^b f(x)dx = F(b) - F(a) = \mathbf{P}\{a < X < b\},$$

where we notice again that the probability in the right-hand side also equals $\mathbf{P}\{a \leq X < b\}$, $\mathbf{P}\{a < X \leq b\}$, and $\mathbf{P}\{a \leq X \leq b\}$.

**Probability density
function**

$$f(x) = F'(x)$$

$$\mathbf{P}\{a < X < b\} = \int_a^b f(x)dx$$

Thus, probabilities can be calculated by integrating a density over the given sets. Furthermore, the integral $\int_a^b f(x)dx$ equals the area below the density curve between the points a and b . Therefore, geometrically, probabilities are represented by *areas* (Figure 4.1). Substituting $a = -\infty$ and $b = +\infty$, we obtain

$$\int_{-\infty}^b f(x)dx = \mathbf{P}\{-\infty < X < b\} = F(b) \quad \text{and} \quad \int_{-\infty}^{+\infty} f(x)dx = \mathbf{P}\{-\infty < X < +\infty\} = 1.$$

That is, the total area below the density curve equals 1.

Looking at Figure 4.1, we can see why $P(x) = 0$ for all continuous random variables. That is because

$$P(x) = \mathbf{P}\{x \leq X \leq x\} = \int_x^x f = 0.$$

Geometrically, it is the area below the density curve, where two sides of the region collapse into one.

Example 4.1. The lifetime, in years, of some electronic component is a continuous random variable with the density

$$f(x) = \begin{cases} \frac{k}{x^3} & \text{for } x \geq 1 \\ 0 & \text{for } x < 1. \end{cases}$$

Find k , draw a graph of the cdf $F(x)$, and compute the probability for the lifetime to exceed 5 years.

Solution. Find k from the condition $\int f(x)dx = 1$:

$$\int_{-\infty}^{+\infty} f(x)dx = \int_1^{+\infty} \frac{k}{x^3}dx = -\frac{k}{2x^2} \Big|_{x=1}^{+\infty} = \frac{k}{2} = 1.$$

Hence, $k = 2$. Integrating the density, we get the cdf,

$$F(x) = \int_{-\infty}^x f(y)dy = \int_1^x \frac{2}{y^3}dy = -\frac{1}{y^2} \Big|_{y=1}^x = 1 - \frac{1}{x^2}$$

for $x > 1$. Its graph is shown in Figure 4.2.

Next, compute the probability for the lifetime to exceed 5 years,

$$P\{X > 5\} = 1 - F(5) = 1 - \left(1 - \frac{1}{5^2}\right) = 0.04.$$

We can also obtain this probability by integrating the density,

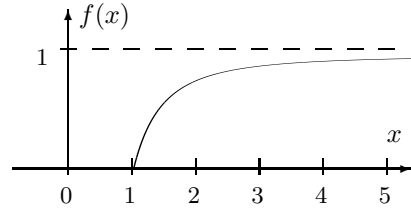


FIGURE 4.2: Cdf for Example 4.1.

$$P\{X > 5\} = \int_5^{+\infty} f(x)dx = \int_5^{+\infty} \frac{2}{x^3}dx = -\frac{1}{x^2} \Big|_{x=5}^{+\infty} = \frac{1}{25} = 0.04.$$

◇

Distribution	Discrete	Continuous
Definition	$P(x) = P\{X = x\}$ (pmf)	$f(x) = F'(x)$ (pdf)
Computing probabilities	$P\{X \in A\} = \sum_{x \in A} P(x)$	$P\{X \in A\} = \int_A f(x)dx$
Cumulative distribution function	$F(x) = P\{X \leq x\} = \sum_{y \leq x} P(y)$	$F(x) = P\{X \leq x\} = \int_{-\infty}^x f(y)dy$
Total probability	$\sum_x P(x) = 1$	$\int_{-\infty}^{\infty} f(x)dx = 1$

TABLE 4.1: Pmf $P(x)$ versus pdf $f(x)$.

Analogy: pmf versus pdf

The role of a density for continuous distributions is very similar to the role of the probability mass function for discrete distributions. Most vital concepts can be translated from the discrete case to the continuous case by replacing pmf $P(x)$ with pdf $f(x)$ and integrating instead of summing, as in Table 4.1.

Joint and marginal densities

DEFINITION 4.2

For a vector of random variables, the **joint cumulative distribution function** is defined as

$$F_{(X,Y)}(x, y) = \mathbf{P} \{X \leq x \cap Y \leq y\}.$$

The **joint density** is the *mixed derivative* of the joint cdf,

$$f_{(X,Y)}(x, y) = \frac{\partial^2}{\partial x \partial y} F_{(X,Y)}(x, y).$$

Similarly to the discrete case, a marginal density of X or Y can be obtained by integrating out the other variable. Variables X and Y are *independent* if their joint density factors into the product of marginal densities. Probabilities about X and Y can be computed by integrating the joint density over the corresponding set of vector values $(x, y) \in \mathbb{R}^2$. This is also analogous to the discrete case; see Table 4.2.

These concepts are directly extended to three or more variables.

Expectation and variance

Continuing our analogy with the discrete case, *expectation* of a continuous variable is also

Distribution	Discrete	Continuous
Marginal distributions	$P(x) = \sum_y P(x, y)$ $P(y) = \sum_x P(x, y)$	$f(x) = \int f(x, y) dy$ $f(y) = \int f(x, y) dx$
Independence	$P(x, y) = P(x)P(y)$	$f(x, y) = f(x)f(y)$
Computing probabilities	$\mathbf{P} \{(X, Y) \in A\}$ $= \sum_{(x,y) \in A} P(x, y)$	$\mathbf{P} \{(X, Y) \in A\}$ $= \iint_{(x,y) \in A} f(x, y) dx dy$

TABLE 4.2: Joint and marginal distributions in discrete and continuous cases.

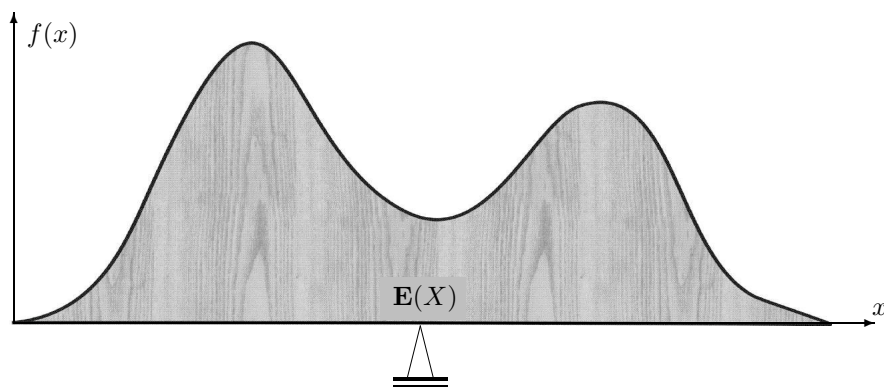


FIGURE 4.3: Expectation of a continuous variable as a center of gravity.

defined as a center of gravity,

$$\mu = \mathbf{E}(X) = \int x f(x) dx$$

(compare with the discrete case on p. 48, Figure 3.3). This time, if the entire region below the density curve is cut from a piece of wood, then it will be balanced at a point with coordinate $\mathbf{E}(X)$, as shown in Figure 4.3.

Variance, standard deviation, covariance, and correlation of continuous variables are defined similarly to the discrete case, see Table 4.3. All the properties in (3.5), (3.7), and (3.8) extend to the continuous distributions. In calculations, don't forget to replace a pmf with a pdf, and a summation with an integral.

Example 4.2. A random variable X in Example 4.1 has density

$$f(x) = 2x^{-3} \text{ for } x \geq 1.$$

Its expectation equals

$$\mu = \mathbf{E}(X) = \int x f(x) dx = \int_1^{\infty} 2x^{-2} dx = -2x^{-1} \Big|_1^{\infty} = 2.$$

Discrete	Continuous
$\mathbf{E}(X) = \sum_x x P(x)$ $\text{Var}(X) = \mathbf{E}(X - \mu)^2$ $= \sum_x (x - \mu)^2 P(x)$ $= \sum_x x^2 P(x) - \mu^2$ $\text{Cov}(X, Y) = \mathbf{E}(X - \mu_X)(Y - \mu_Y)$ $= \sum_x \sum_y (x - \mu_X)(y - \mu_Y) P(x, y)$ $= \sum_x \sum_y (xy) P(x, y) - \mu_x \mu_y$	$\mathbf{E}(X) = \int x f(x) dx$ $\text{Var}(X) = \mathbf{E}(X - \mu)^2$ $= \int (x - \mu)^2 f(x) dx$ $= \int x^2 f(x) dx - \mu^2$ $\text{Cov}(X, Y) = \mathbf{E}(X - \mu_X)(Y - \mu_Y)$ $= \iint (x - \mu_X)(y - \mu_Y) f(x, y) dx dy$ $= \iint (xy) f(x, y) dx dy - \mu_x \mu_y$

TABLE 4.3: Moments for discrete and continuous distributions.

Computing its variance, we run into a “surprise,”

$$\sigma^2 = \text{Var}(X) = \int x^2 f(x) dx - \mu^2 = \int_1^\infty 2x^{-1} dx - 4 = 2 \ln x|_1^\infty - 4 = +\infty.$$

This variable does not have a finite variance! (Also, see the St. Petersburg paradox discussion on p. 62.) $\diamond$

4.2 Families of continuous distributions

As in the discrete case, varieties of phenomena can be described by relatively few families of continuous distributions. Here, we shall discuss Uniform, Exponential, Gamma, and Normal families, adding Student's t , Pearson's χ^2 , and Fisher's F distributions in later chapters.

4.2.1 Uniform distribution

Uniform distribution plays a unique role in stochastic modeling. As we shall see in Chapter 5, a random variable with any thinkable distribution can be generated from a Uniform random variable. Many computer languages and software are equipped with a random number generator that produces Uniform random variables. Users can convert them into variables with desired distributions and use for computer simulation of various events and processes.

Also, Uniform distribution is used in any situation when a value is picked “at random” from a given interval; that is, without any preference to lower, higher, or medium values. For example, locations of errors in a program, birthdays throughout a year, and many continuous random variables modulo 1, modulo 0.1, 0.01, etc., are uniformly distributed over their corresponding intervals.

To give equal preference to all values, the Uniform distribution has a *constant* density (Figure 4.4). On the interval (a, b) , its density equals

$$f(x) = \frac{1}{b-a}, \quad a < x < b,$$

because the rectangular area below the density graph must equal 1.

For the same reason, $|b-a|$ has to be a finite number. There does not exist a Uniform distribution on the entire real line. In other words, if you are asked to choose a random number from $(-\infty, +\infty)$, you cannot do it uniformly.

The Uniform property

For any $h > 0$ and $t \in [a, b-h]$, the probability

$$P\{t < X < t+h\} = \int_t^{t+h} \frac{1}{b-a} dx = \frac{h}{b-a}$$

is *independent* of t . This is the *Uniform property*: the probability is only determined by the length of the interval, but not by its location.

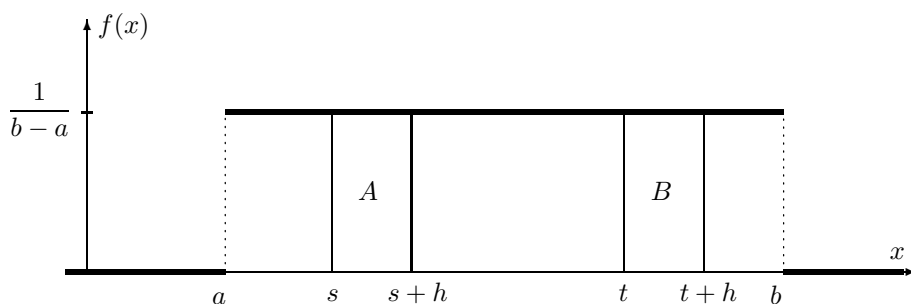


FIGURE 4.4: The Uniform density and the Uniform property.

Example 4.3. In Figure 4.4, rectangles A and B have the same area, showing that $P\{s < X < s+h\} = P\{t < X < t+h\}$. $\diamond$

Example 4.4. If a flight scheduled to arrive at 5 pm actually arrives at a Uniformly distributed time between 4:50 and 5:10, then it is equally likely to arrive before 5 pm and after 5 pm, equally likely before 4:55 and after 5:05, etc. $\diamond$

Standard Uniform distribution

The Uniform distribution with $a = 0$ and $b = 1$ is called *Standard Uniform distribution*. The Standard Uniform density is $f(x) = 1$ for $0 < x < 1$. Most random number generators return a Standard Uniform random variable.

All the Uniform distributions are related in the following way. If X is a $\text{Uniform}(a, b)$ random variable, then

$$Y = \frac{X - a}{b - a}$$

is Standard Uniform. Likewise, if Y is Standard Uniform, then

$$X = a + (b - a)Y$$

is $\text{Uniform}(a, b)$. Check that $X \in (a, b)$ if and only if $Y \in (0, 1)$.

A number of other families of distributions have a “standard” member. Typically, a simple transformation converts a standard random variable into a non-standard one, and vice versa.

Expectation and variance

For a Standard Uniform variable Y ,

$$\mathbf{E}(Y) = \int_0^1 y f(y) dy = \int_0^1 y dy = \frac{1}{2}$$

and

$$\text{Var}(Y) = \mathbf{E}(Y^2) - \mathbf{E}^2(Y) = \int_0^1 y^2 dy - \left(\frac{1}{2}\right)^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}.$$

Now, consider the general case. Let $X = a + (b - a)Y$ which has a $\text{Uniform}(a, b)$ distribution. By the properties of expectations and variances in (3.5) and (3.7),

$$\mathbf{E}(X) = \mathbf{E}\{a + (b - a)Y\} = a + (b - a)\mathbf{E}(Y) = a + \frac{b - a}{2} = \frac{a + b}{2}$$

and

$$\text{Var}(X) = \text{Var}\{a + (b - a)Y\} = (b - a)^2 \text{Var}(Y) = \frac{(b - a)^2}{12}.$$

The expectation is precisely the middle of the interval $[a, b]$. Giving no preference to left or right sides, this agrees with the Uniform property and with the physical meaning of $\mathbf{E}(X)$ as a center of gravity.

**Uniform
distribution**

(a, b)	=	range of values
$f(x)$	=	$\frac{1}{b - a}, \quad a < x < b$
$\mathbf{E}(X)$	=	$\frac{a + b}{2}$
$\text{Var}(X)$	=	$\frac{(b - a)^2}{12}$

4.2.2 Exponential distribution

Exponential distribution is often used to model *time*: waiting time, interarrival time, hardware lifetime, failure time, time between telephone calls, etc. As we shall see below, in a sequence of rare events, when the number of events is Poisson, the time between events is Exponential.

Exponential distribution has density

$$f(x) = \lambda e^{-\lambda x} \quad \text{for } x > 0. \quad (4.1)$$

With this density, we compute the Exponential cdf, mean, and variance as

$$F(x) = \int_0^x f(t)dt = \int_0^x \lambda e^{-\lambda t} dt = 1 - e^{-\lambda x} \quad (x > 0), \quad (4.2)$$

$$\mathbf{E}(X) = \int_0^\infty t f(t)dt = \int_0^\infty t \lambda e^{-\lambda t} dt = \frac{1}{\lambda} \quad \left(\begin{array}{c} \text{integrating} \\ \text{by parts} \end{array} \right), \quad (4.3)$$

$$\begin{aligned} \text{Var}(X) &= \int_0^\infty t^2 f(t)dt - \mathbf{E}^2(X) \\ &= \int_0^\infty t^2 \lambda e^{-\lambda t} dt - \left(\frac{1}{\lambda}\right)^2 \quad (\text{by parts twice}) \\ &= \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}. \end{aligned} \quad (4.4)$$

The quantity λ is a parameter of Exponential distribution, and its meaning is clear from

$\mathbf{E}(X) = 1/\lambda$. If X is time, measured in minutes, then λ is a frequency, measured in min^{-1} . For example, if arrivals occur every half a minute, on the average, then $\mathbf{E}(X) = 0.5$ and $\lambda = 2$, saying that they occur with a frequency (arrival rate) of 2 arrivals per minute. This λ has the same meaning as the parameter of Poisson distribution in Section 3.4.5.

Times between rare events are Exponential

What makes Exponential distribution a good model for interarrival times? Apparently, this is not only experimental, but also a mathematical fact.

As in Section 3.4.5, consider the sequence of rare events, where the number of occurrences during time t has Poisson distribution with a parameter proportional to t . This process is rigorously defined in Section 6.3.2 where we call it *Poisson process*.

Event “the time T until the next event is greater than t ” can be rephrased as “zero events occur by the time t ,” and further, as “ $X = 0$,” where X is the number of events during the time interval $[0, t]$. This X has Poisson distribution with parameter λt . It equals 0 with probability

$$P_X(0) = e^{-\lambda t} \frac{(\lambda t)^0}{0!} = e^{-\lambda t}.$$

Then we can compute the cdf of T as

$$F_T(t) = 1 - \mathbf{P}\{T > t\} = 1 - \mathbf{P}\{X = 0\} = 1 - e^{-\lambda t}, \quad (4.5)$$

and here we recognize the Exponential cdf. Therefore, the time until the next arrival has Exponential distribution.

Example 4.5. Jobs are sent to a printer at an average rate of 3 jobs per hour.

- (a) What is the expected time between jobs?
- (b) What is the probability that the next job is sent within 5 minutes?

Solution. Job arrivals represent rare events, thus the time T between them is Exponential with the given parameter $\lambda = 3 \text{ hrs}^{-1}$ (jobs per hour).

- (a) $\mathbf{E}(T) = 1/\lambda = 1/3$ hours or 20 minutes between jobs;
- (b) Convert to the same measurement unit: 5 min = $(1/12)$ hrs. Then,

$$\mathbf{P}\{T < 1/12 \text{ hrs}\} = F(1/12) = 1 - e^{-\lambda(1/12)} = 1 - e^{-1/4} = \underline{0.2212}.$$

◇

Memoryless property

It is said that “Exponential variables lose memory.” What does it mean?

Suppose that an Exponential variable T represents waiting time. Memoryless property means that the fact of having waited for t minutes gets “forgotten,” and it does not affect the future waiting time. Regardless of the event $T > t$, when the total waiting time exceeds

t , the remaining waiting time still has Exponential distribution with the same parameter. Mathematically,

$$\mathbf{P}\{T > t + x \mid T > t\} = \mathbf{P}\{T > x\} \quad \text{for } t, x > 0. \quad (4.6)$$

In this formula, t is the already elapsed portion of waiting time, and x is the additional, remaining time.

PROOF: From (4.2), $\mathbf{P}\{T > x\} = e^{-\lambda x}$. Also, by the formula (2.7) for conditional probability,

$$\mathbf{P}\{T > t + x \mid T > t\} = \frac{\mathbf{P}\{T > t + x \cap T > t\}}{\mathbf{P}\{T > t\}} = \frac{\mathbf{P}\{T > t + x\}}{\mathbf{P}\{T > t\}} = \frac{e^{-\lambda(t+x)}}{e^{-\lambda t}} = e^{-\lambda x}.$$

□

This property is unique for Exponential distribution. No other continuous variable $X \in (0, \infty)$ is memoryless. Among discrete variables, such a property belongs to *Geometric distribution* (Exercise 4.34).

In a sense, Geometric distribution is a discrete analogue of Exponential. Connection between these two families of distributions will become clear in Section 5.2.3.

**Exponential
distribution**

λ	$=$	frequency parameter, the number of events per time unit
$f(x)$	$=$	$\lambda e^{-\lambda x}, \quad x > 0$
$\mathbf{E}(X)$	$=$	$\frac{1}{\lambda}$
$\text{Var}(X)$	$=$	$\frac{1}{\lambda^2}$

4.2.3 Gamma distribution

When a certain procedure consists of α independent steps, and each step takes $\text{Exponential}(\lambda)$ amount of time, then the total time has *Gamma distribution* with parameters α and λ .

Thus, Gamma distribution can be widely used for the total time of a multistage scheme, for example, related to downloading or installing a number of files. In a process of rare events, with Exponential times between any two consecutive events, the time of the α -th event has Gamma distribution because it consists of α independent Exponential times.

Example 4.6 (INTERNET PROMOTIONS). Users visit a certain internet site at the average rate of 12 hits per minute. Every sixth visitor receives some promotion that comes in a form of a flashing banner. Then the time between consecutive promotions has Gamma distribution with parameters $\alpha = 6$ and $\lambda = 12$. ◇

Having two parameters, Gamma distribution family offers a variety of models for positive

random variables. Besides the case when a Gamma variable represents a sum of independent Exponential variables, Gamma distribution is often used for the amount of money being paid, amount of a commodity being used (gas, electricity, etc.), a loss incurred by some accident, etc.

Gamma distribution has a density

$$f(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, \quad x > 0. \quad (4.7)$$

The denominator contains a *Gamma function*, Section 12.3.5. With certain techniques, this density can be mathematically derived for integer α by representing a Gamma variable X as a sum of Exponential variables each having a density (4.1).

In fact, α can take any positive value, not necessarily integer. With different α , the Gamma density takes different shapes (Figure 4.5). For this reason, α is called a *shape parameter*.

Notice two important special cases of a Gamma distribution. When $\alpha = 1$, the Gamma distribution becomes *Exponential*. This can be seen comparing (4.7) and (4.1) for $\alpha = 1$. Another special case with $\lambda = 1/2$ and any $\alpha > 0$ results in a so-called *Chi-square distribution* with (2α) degrees of freedom, which we discuss in Section 9.5.1 and use a lot in Chapter 10.

$\begin{aligned} \text{Gamma}(1, \lambda) &= \text{Exponential}(\lambda) \\ \text{Gamma}(\alpha, 1/2) &= \text{Chi-square}(2\alpha) \end{aligned}$
--

Expectation, variance, and some useful integration remarks

Gamma cdf has the form

$$F(t) = \int_0^t f(x) dx = \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^t x^{\alpha-1} e^{-\lambda x} dx. \quad (4.8)$$

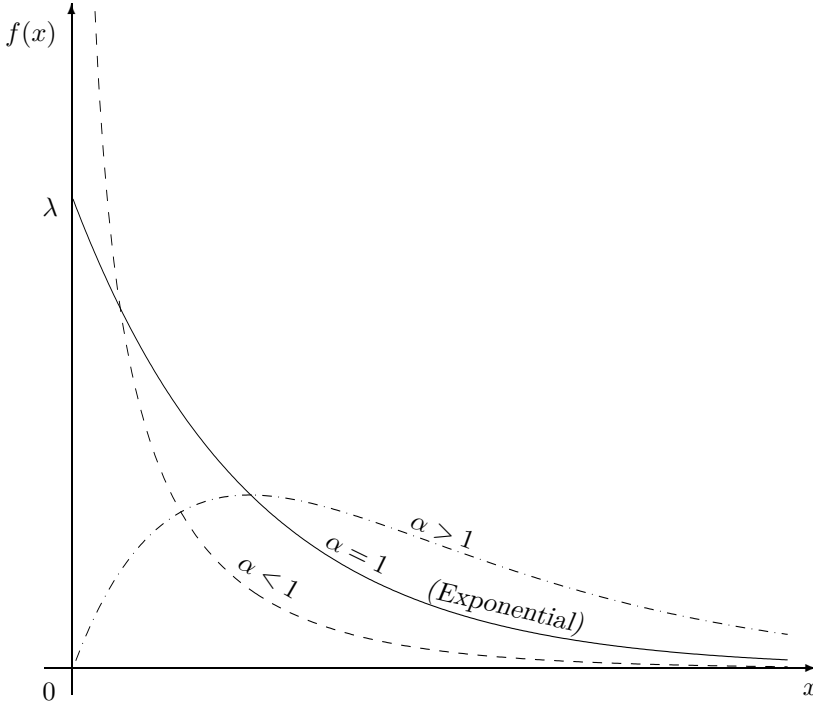
This expression, related to a so-called *incomplete Gamma function*, does not simplify, and thus, computing probabilities is not always trivial. Let us offer several computational short-cuts.

First, let us notice that $\int_0^\infty f(x) dx = 1$ for Gamma and all the other densities. Then, integrating (4.7) from 0 to ∞ , we obtain that

$\int_0^\infty x^{\alpha-1} e^{-\lambda x} dx = \frac{\Gamma(\alpha)}{\lambda^\alpha} \quad \text{for any } \alpha > 0 \text{ and } \lambda > 0$	(4.9)
--	-------

Substituting $\alpha + 1$ and $\alpha + 2$ in place of α , we get for a Gamma variable X ,

$$\mathbf{E}(X) = \int x f(x) dx = \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty x^\alpha e^{-\lambda x} dx = \frac{\lambda^\alpha}{\Gamma(\alpha)} \cdot \frac{\Gamma(\alpha + 1)}{\lambda^{\alpha+1}} = \frac{\alpha}{\lambda} \quad (4.10)$$

FIGURE 4.5: Gamma densities with different shape parameters α .

(using the equality $\Gamma(t+1) = t\Gamma(t)$ that holds for all $t > 0$),

$$\mathbf{E}(X^2) = \int x^2 f(x) dx = \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty x^{\alpha+1} e^{-\lambda x} dx = \frac{\lambda^\alpha}{\Gamma(\alpha)} \cdot \frac{\Gamma(\alpha+2)}{\lambda^{\alpha+2}} = \frac{(\alpha+1)\alpha}{\lambda^2},$$

and therefore,

$$\text{Var}(X) = \mathbf{E}(X^2) - \mathbf{E}^2(X) = \frac{(\alpha+1)\alpha - \alpha^2}{\lambda^2} = \frac{\alpha}{\lambda^2}. \quad (4.11)$$

For $\alpha = 1$, this agrees with (4.3) and (4.4). Moreover, for any integer α , (4.10) and (4.11) can be obtained directly from (4.3) and (4.4) by representing a Gamma variable X as a sum of independent $\text{Exponential}(\lambda)$ variables $X_1, \dots, X_\alpha$,

$$\mathbf{E}(X) = \mathbf{E}(X_1 + \dots + X_\alpha) = \mathbf{E}(X_1) + \dots + \mathbf{E}(X_\alpha) = \alpha \left(\frac{1}{\lambda} \right),$$

$$\text{Var}(X) = \text{Var}(X_1 + \dots + X_\alpha) = \text{Var}(X_1) + \dots + \text{Var}(X_\alpha) = \alpha \left(\frac{1}{\lambda^2} \right).$$

**Gamma
distribution**

α	$=$ shape parameter	(4.12)
λ	$=$ frequency parameter	
$f(x)$	$= \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, \quad x > 0$	
$\mathbf{E}(X)$	$= \frac{\alpha}{\lambda}$	
$\text{Var}(X)$	$= \frac{\alpha}{\lambda^2}$	

Example 4.7 (TOTAL COMPILATION TIME). Compilation of a computer program consists of 3 blocks that are processed sequentially, one after another. Each block takes Exponential time with the mean of 5 minutes, independently of other blocks.

- (a) Compute the expectation and variance of the total compilation time.
 (b) Compute the probability for the entire program to be compiled in less than 12 minutes.

Solution. The total time T is a sum of three independent Exponential times, therefore, it has Gamma distribution with $\alpha = 3$. The frequency parameter λ equals $(1/5) \text{ min}^{-1}$ because the Exponential compilation time of each block has expectation $1/\lambda = 5 \text{ min}$.

- (a) For a Gamma random variable T with $\alpha = 3$ and $\lambda = 1/5$,

$$\mathbf{E}(T) = \frac{3}{1/5} = 15 \text{ (min)} \quad \text{and} \quad \text{Var}(T) = \frac{3}{(1/5)^2} = 75 \text{ (min}^2\text{)}.$$

- (b) A direct solution involves two rounds of integration by parts,

$$\begin{aligned} P\{T < 12\} &= \int_0^{12} f(t)dt = \frac{(1/5)^3}{\Gamma(3)} \int_0^{12} t^2 e^{-t/5} dt \\ &= \frac{(1/5)^3}{2!} \left(-5t^2 e^{-t/5} \Big|_{t=0}^{t=12} + \int_0^{12} 10te^{-t/5} dt \right) \\ &= \frac{1/125}{2} \left(-5t^2 e^{-t/5} - 50te^{-t/5} \Big|_{t=0}^{t=12} + \int_0^{12} 50e^{-t/5} dt \right) \\ &= \frac{1}{250} \left(-5t^2 e^{-t/5} - 50te^{-t/5} - 250e^{-t/5} \right) \Big|_{t=0}^{t=12} \\ &= 1 - e^{-2.4} - 2.4e^{-2.4} - 2.88e^{-2.4} = \underline{0.4303}. \end{aligned} \tag{4.13}$$

A much shorter way is to apply the Gamma-Poisson formula below (Example 4.8). $\diamond$

Gamma-Poisson formula

Computation of Gamma probabilities can be significantly simplified by thinking of a Gamma variable as the time between some rare events. In particular, one can avoid lengthy integration by parts, as in Example 4.7, and use Poisson distribution instead.

Indeed, let T be a Gamma variable with an integer parameter α and some positive λ . This is a distribution of the time of the α -th rare event. Then, the event $\{T > t\}$ means that the α -th rare event occurs after the moment t , and therefore, *fewer than α rare events occur before the time t* . We see that

$$\{T > t\} = \{X < \alpha\},$$

where X is the number of events that occur before the time t . This number of rare events X has Poisson distribution with parameter (λt) ; therefore, the probability

$$P\{T > t\} = P\{X < \alpha\}$$

and the probability of a complement

$$P\{T \leq t\} = P\{X \geq \alpha\}$$

can both be computed using the Poisson distribution of X .

**Gamma-Poisson
formula**

For a $\text{Gamma}(\alpha, \lambda)$ variable T
and a $\text{Poisson}(\lambda t)$ variable X ,

$$\mathbf{P}\{T > t\} = \mathbf{P}\{X < \alpha\} \quad (4.14)$$

$$\mathbf{P}\{T \leq t\} = \mathbf{P}\{X \geq \alpha\}$$

Remark: Recall that $\mathbf{P}\{T > t\} = \mathbf{P}\{T \geq t\}$ and $\mathbf{P}\{T < t\} = \mathbf{P}\{T \leq t\}$ for a Gamma variable T , because it is continuous. Hence, (4.14) can also be used for the computation of $\mathbf{P}\{T \geq t\}$ and $\mathbf{P}\{T < t\}$. Conversely, the probability of $\{X = \alpha\}$ cannot be neglected for the Poisson (discrete!) variable X , thus the signs in the right-hand sides of (4.14) cannot be altered.

Example 4.8 (TOTAL COMPILATION TIME, CONTINUED). Here is an alternative solution to Example 4.7(b). According to the Gamma-Poisson formula with $\alpha = 3$, $\lambda = 1/5$, and $t = 12$,

$$\mathbf{P}\{T < 12\} = \mathbf{P}\{X \geq 3\} = 1 - F(2) = 1 - 0.5697 = \underline{0.430}$$

from Table A3 for the Poisson distribution of X with parameter $\lambda t = 2.4$.

Furthermore, we notice that the four-term mathematical expression that we obtained in (4.13) after integrating by parts represents precisely

$$\mathbf{P}\{X \geq 3\} = 1 - P(0) - P(1) - P(2).$$

◇

Example 4.9. Lifetimes of computer memory chips have Gamma distribution with expectation $\mu = 12$ years and standard deviation $\sigma = 4$ years. What is the probability that such a chip has a lifetime between 8 and 10 years?

Solution.

STEP 1, PARAMETERS. From the given data, compute parameters of this Gamma distribution. Using (4.12), obtain a system of two equations and solve them for α and λ ,

$$\begin{cases} \mu &= \alpha/\lambda \\ \sigma^2 &= \alpha/\lambda^2 \end{cases} \Rightarrow \begin{cases} \alpha &= \mu^2/\sigma^2 = (12/4)^2 = 9, \\ \lambda &= \mu/\sigma^2 = 12/4^2 = 0.75. \end{cases}$$

STEP 2, PROBABILITY. We can now compute the probability,

$$\mathbf{P}\{8 < T < 10\} = F_T(10) - F_T(8). \quad (4.15)$$

For each term in (4.15), we use the Gamma-Poisson formula with $\alpha = 9$, $\lambda = 0.75$, and $t = 8, 10$,

$$F_T(10) = \mathbf{P}\{T \leq 10\} = \mathbf{P}\{X \geq 9\} = 1 - F_X(8) = 1 - 0.662 = 0.338$$

from Table A3 for a Poisson variable X with parameter $\lambda t = (0.75)(10) = 7.5$;

$$F_T(8) = \mathbf{P}\{T \leq 8\} = \mathbf{P}\{X \geq 9\} = 1 - F_X(8) = 1 - 0.847 = 0.153$$

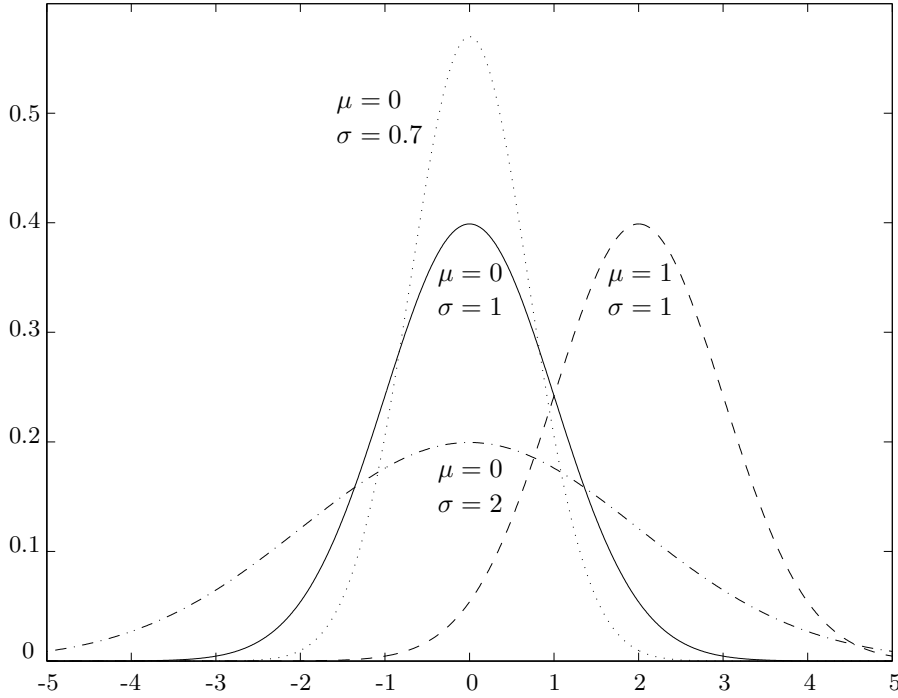


FIGURE 4.6: Normal densities with different location and scale parameters.

from the same table, this time with parameter $\lambda t = (0.75)(8) = 6$. Then,

$$P\{8 < T < 10\} = 0.338 - 0.153 = \underline{0.185}.$$

◇

4.2.4 Normal distribution

Normal distribution plays a vital role in Probability and Statistics, mostly because of the Central Limit Theorem, according to which sums and averages often have approximately Normal distribution. Due to this fact, various fluctuations and measurement errors that consist of accumulated number of small terms appear normally distributed.

Remark: As said by a French mathematician *Jules Henri Poincaré*, “Everyone believes in the Normal law of errors, the experimenters because they think it is a mathematical theorem, the mathematicians because they think it is an experimental fact.”

Besides sums, averages, and errors, Normal distribution is often found to be a good model for physical variables like weight, height, temperature, voltage, pollution level, and for instance, household incomes or student grades.

Normal distribution has a density

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}, \quad -\infty < x < +\infty,$$

where parameters μ and σ have a simple meaning of the expectation $\mathbf{E}(X)$ and the standard deviation $\text{Std}(X)$. This density is known as the bell-shaped curve, symmetric and centered at μ , its spread being controlled by σ . As seen in Figure 4.6, changing μ shifts the curve to the left or to the right without affecting its shape, while changing σ makes it more concentrated or more flat. Often μ and σ are called *location* and *scale* parameters.

**Normal
distribution**

μ	=	expectation, location parameter
σ	=	standard deviation, scale parameter
$f(x)$	=	$\frac{1}{\sigma\sqrt{2\pi}} \exp\left\{\frac{-(x-\mu)^2}{2\sigma^2}\right\}, -\infty < x < \infty$
$\mathbf{E}(X)$	=	μ
$\text{Var}(X)$	=	σ^2

Standard Normal distribution

DEFINITION 4.3

Normal distribution with “standard parameters” $\mu = 0$ and $\sigma = 1$ is called **Standard Normal distribution**.

NOTATION

Z	=	Standard Normal random variable
$\phi(x)$	=	$\frac{1}{\sqrt{2\pi}} e^{-x^2/2}$, Standard Normal pdf
$\Phi(x)$	=	$\int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz$, Standard Normal cdf

A Standard Normal variable, usually denoted by Z , can be obtained from a non-standard $\text{Normal}(\mu, \sigma)$ random variable X by *standardizing*, that is, subtracting the mean and dividing by the standard deviation,

$$Z = \frac{X - \mu}{\sigma}. \quad (4.16)$$

Unstandardizing Z , we can reconstruct the initial variable X ,

$$X = \mu + \sigma Z. \quad (4.17)$$

Using these transformations, any Normal random variable can be obtained from a Standard Normal variable Z ; therefore, we need a table of Standard Normal Distribution only (Table A4).

To find $\Phi(z)$ from Table A4, we locate a row with the first two digits of z and a column

with the third digit of z and read the probability $\Phi(z)$ at their intersection. Notice that $\Phi(z) \approx 0$ (is “practically” zero) for all $z < -3.9$, and $\Phi(z) \approx 1$ (is “practically” one) for all $z > 3.9$.

Example 4.10 (COMPUTING STANDARD NORMAL PROBABILITIES). For a Standard Normal random variable Z ,

$$\begin{aligned} P\{Z < 1.35\} &= \Phi(1.35) = 0.9115 \\ P\{Z > 1.35\} &= 1 - \Phi(1.35) = 0.0885 \\ P\{-0.77 < Z < 1.35\} &= \Phi(1.35) - \Phi(-0.77) = 0.9115 - 0.2206 = 0.6909. \end{aligned}$$

according to Table A4. Notice that $P\{Z < -1.35\} = 0.0885 = P\{Z > 1.35\}$, which is explained by the symmetry of the Standard Normal density in Figure 4.6. Due to this symmetry, “the left tail,” or the area to the left of (-1.35) equals “the right tail,” or the area to the right of 1.35 . $\diamond$

In fact, the symmetry of the Normal density, mentioned in this example, allows to obtain the first part of Table A4 on p. 417 directly from the second part,

$$\Phi(-z) = 1 - \Phi(z) \quad \text{for } -\infty < z < +\infty.$$

To compute probabilities about an arbitrary Normal random variable X , we have to standardize it first, as in (4.16), then use Table A4.

Example 4.11 (COMPUTING NON-STANDARD NORMAL PROBABILITIES). Suppose that the average household income in some country is 900 coins, and the standard deviation is 200 coins. Assuming the Normal distribution of incomes, compute the proportion of “the middle class,” whose income is between 600 and 1200 coins.

Solution. Standardize and use Table A4. For a Normal($\mu = 900$, $\sigma = 200$) variable X ,

$$\begin{aligned} P\{600 < X < 1200\} &= P\left\{\frac{600 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{1200 - \mu}{\sigma}\right\} \\ &= P\left\{\frac{600 - 900}{200} < Z < \frac{1200 - 900}{200}\right\} = P\{-1.5 < Z < 1.5\} \\ &= \Phi(1.5) - \Phi(-1.5) = 0.9332 - 0.0668 = \underline{0.8664}. \end{aligned}$$

$\diamond$

So far, we were computing probabilities of clearly defined events. These are *direct* problems. A number of applications require solution of an *inverse problem*, that is, finding a value of x given the corresponding probability.

Example 4.12 (INVERSE PROBLEM). The government of the country in Example 4.11 decides to issue food stamps to the poorest 3% of households. Below what income will families receive food stamps?

Solution. We need to find such income x that $P\{X < x\} = 3\% = 0.03$. This is an equation that can be solved in terms of x . Again, we standardize first, then use the table:

$$P\{X < x\} = P\left\{Z < \frac{x - \mu}{\sigma}\right\} = \Phi\left(\frac{x - \mu}{\sigma}\right) = 0.03,$$

from where

$$x = \mu + \sigma \Phi^{-1}(0.03).$$

In Table A4, we have to find the probability, the *table entry* of 0.03. We see that $\Phi(-1.88) \approx 0.03$. Therefore, $\Phi^{-1}(0.03) = -1.88$, and

$$x = \mu + \sigma(-1.88) = 900 + (200)(-1.88) = \underline{524} \text{ (coins)}$$

is the answer. In the literature, the value $\Phi^{-1}(\alpha)$ is often denoted by $z_{1-\alpha}$. $\diamond$

As seen in this example, in order to solve an inverse problem, we use the table first, then *unstandardize*, as in (4.17), and find the required value of x .

4.3 Central Limit Theorem

We now turn our attention to *sums* of random variables,

$$S_n = X_1 + \dots + X_n,$$

that appear in many applications. Let $\mu = \mathbf{E}(X_i)$ and $\sigma = \text{Std}(X_i)$ for all $i = 1, \dots, n$. How does S_n behave for large n ?

MATLAB DEMO. The following MATLAB code is a good illustration to the behavior of partial sums S_n .

```
S(1)=0;
for n=2:1000; S(n)=S(n-1)+randn; end; % nth partial sum
n=1:1000; comet(n,S); pause(3); % Behavior of S(n)
comet(n,S./n); pause(3); % Behavior of S(n)/n
comet(n,S./sqrt(n)); pause(3); % Behavior of S(n)/√n
```

Apparently (MATLAB users can see it from the obtained graphs),

- The *pure sum* S_n *diverges*. In fact, this should be anticipated because

$$\text{Var}(S_n) = n\sigma^2 \rightarrow \infty,$$

so that variability of S_n grows unboundedly as n goes to infinity.

- The *average* S_n/n *converges*. Indeed, in this case, we have

$$\text{Var}(S_n/n) = \text{Var}(S_n)/n^2 = n\sigma^2/n^2 = \sigma^2/n \rightarrow 0,$$

so that variability of (S_n/n) vanishes as $n \rightarrow \infty$.

- An interesting normalization factor is $1/\sqrt{n}$. For $\mu = 0$, we can see from MATLAB simulations that $S_n/\sqrt{n}$ *neither diverges nor converges!* It does not tend to leave 0, but it does not converge to 0 either. Rather, it behaves like some random variable. The following theorem states that this variable has approximately Normal distribution for large n .

Theorem 1 (CENTRAL LIMIT THEOREM) *Let $X_1, X_2, \dots$ be independent random variables with the same expectation $\mu = \mathbf{E}(X_i)$ and the same standard deviation $\sigma = \text{Std}(X_i)$, and let*

$$S_n = \sum_{i=1}^n X_i = X_1 + \dots + X_n.$$

As $n \rightarrow \infty$, the standardized sum

$$Z_n = \frac{S_n - \mathbf{E}(S_n)}{\text{Std}(S_n)} = \frac{S_n - n\mu}{\sigma\sqrt{n}}$$

converges in distribution to a Standard Normal random variable, that is,

$$F_{Z_n}(z) = \mathbf{P}\left\{\frac{S_n - n\mu}{\sigma\sqrt{n}} \leq z\right\} \rightarrow \Phi(z) \quad (4.18)$$

for all z .

This theorem is very powerful because it can be applied to random variables $X_1, X_2, \dots$ having virtually any thinkable distribution with finite expectation and variance. As long as n is large (the rule of thumb is $n > 30$), one can use Normal distribution to compute probabilities about S_n .

Theorem 1 is only one basic version of the Central Limit Theorem. Over the last two centuries, it has been extended to large classes of dependent variables and vectors, stochastic processes, and so on.

Example 4.13 (ALLOCATION OF DISK SPACE). A disk has free space of 330 megabytes. Is it likely to be sufficient for 300 independent images, if each image has expected size of 1 megabyte with a standard deviation of 0.5 megabytes?

Solution. We have $n = 300$, $\mu = 1$, $\sigma = 0.5$. The number of images n is large, so the Central Limit Theorem applies to their total size S_n . Then,

$$\begin{aligned} \mathbf{P}\{\text{sufficient space}\} &= \mathbf{P}\{S_n \leq 330\} = \mathbf{P}\left\{\frac{S_n - n\mu}{\sigma\sqrt{n}} \leq \frac{330 - (300)(1)}{0.5\sqrt{300}}\right\} \\ &\approx \Phi(3.46) = 0.9997. \end{aligned}$$

This probability is very high, hence, the available disk space is very likely to be sufficient.
 $\diamond$

In the special case of *Normal* variables $X_1, X_2, \dots$, the distribution of S_n is always Normal, and (4.18) becomes exact equality for arbitrary, even small n .

Example 4.14 (ELEVATOR). You wait for an elevator, whose capacity is 2000 pounds. The elevator comes with ten adult passengers. Suppose your own weight is 150 lbs, and you heard that human weights are normally distributed with the mean of 165 lbs and the standard deviation of 20 lbs. Would you board this elevator or wait for the next one?

Solution. In other words, is overload likely? The probability of an overload equals

$$\begin{aligned} \mathbf{P}\{S_{10} + 150 > 2000\} &= \mathbf{P}\left\{\frac{S_{10} - (10)(165)}{20\sqrt{10}} > \frac{2000 - 150 - (10)(165)}{20\sqrt{10}}\right\} \\ &= 1 - \Phi(3.16) = 0.0008. \end{aligned}$$

So, with probability 0.9992 it is safe to take this elevator. It is now for you to decide.

◇

Among the random variables discussed in Chapters 3 and 4, at least three have a form of S_n :

Binomial variable	=	sum of independent Bernoulli variables
Negative Binomial variable	=	sum of independent Geometric variables
Gamma variable	=	sum of independent Exponential variables

Hence, the Central Limit Theorem applies to all these distributions with sufficiently large n in the case of Binomial, k for Negative Binomial, and α for Gamma variables.

In fact, *Abraham de Moivre* (1667–1754) obtained the first version of the Central Limit Theorem as the approximation of Binomial distribution.

Normal approximation to Binomial distribution

Binomial variables represent a special case of $S_n = X_1 + \dots + X_n$, where all X_i have Bernoulli distribution with some parameter p . We know from Section 3.4.5 that small p allows to approximate Binomial distribution with Poisson, and large p allows such an approximation for the number of failures. For the moderate values of p (say, $0.05 \leq p \leq 0.95$) and for large n , we can use Theorem 1:

$$\text{Binomial}(n, p) \approx \text{Normal} \left(\mu = np, \sigma = \sqrt{np(1-p)} \right) \quad (4.19)$$

MATLAB DEMO. To visualize how Binomial distribution gradually takes the shape of Normal distribution when $n \rightarrow \infty$, you may execute the following MATLAB code that graphs $\text{Binomial}(n, p)$ pmf for increasing values of n .

```
for n=1:2:100; x=0:n;
p=gamma(n+1)./gamma(x+1)./gamma(n-x+1).*(0.3).^x.*(0.7).^(n-x);
plot(x,p);
title('Binomial probabilities for increasing values of n');
pause(1); end;
```

Continuity correction

This correction is needed when we approximate a discrete distribution (Binomial in this case) by a continuous distribution (Normal). Recall that the probability $\mathbf{P}\{X = x\}$ may be positive if X is discrete, whereas it is always 0 for continuous X . Thus, a direct use of (4.19) will always approximate this probability by 0. It is obviously a poor approximation.

This is resolved by introducing a *continuity correction*. Expand the interval by 0.5 units in each direction, then use the Normal approximation. Notice that

$$P_X(x) = \mathbf{P}\{X = x\} = \mathbf{P}\{x - 0.5 < X < x + 0.5\}$$

is true for a Binomial variable X ; therefore, the continuity correction does not change the event and preserves its probability. It makes a difference for the Normal distribution, so every time when we approximate some discrete distribution with some continuous distribution, we should be using a continuity correction. Now it is the probability of an interval instead of one number, and it is not zero.

Example 4.15. A new computer virus attacks a folder consisting of 200 files. Each file gets damaged with probability 0.2 independently of other files. What is the probability that fewer than 50 files get damaged?

Solution. The number X of damaged files has Binomial distribution with $n = 200$, $p = 0.2$, $\mu = np = 40$, and $\sigma = \sqrt{np(1-p)} = 5.657$. Applying the Central Limit Theorem with the continuity correction,

$$\begin{aligned} P\{X < 50\} &= P\{X < 49.5\} = P\left\{\frac{X - 40}{5.657} < \frac{49.5 - 40}{5.657}\right\} \\ &= \Phi(1.68) = \underline{0.9535}. \end{aligned}$$

Notice that the properly applied continuity correction replaces 50 with 49.5, not 50.5. Indeed, we are interested in the event that X is *strictly* less than 50. This includes all values up to 49 and corresponds to the interval $[0, 49]$ that we *expand* to $[0, 49.5]$. In other words, events $\{X < 50\}$ and $\{X < 49.5\}$ are the same; they include the same possible values of X . Events $\{X < 50\}$ and $\{X < 50.5\}$ are different because the former includes $X = 50$, and the latter does not. Replacing $\{X < 50\}$ with $\{X < 50.5\}$ would have changed its probability and would have given a wrong answer. $\diamond$

When a continuous distribution (say, Gamma) is approximated by another continuous distribution (Normal), the continuity correction is not needed. In fact, it would be an error to use it in this case because it would no longer preserve the probability.

Summary and conclusions

Continuous distributions are used to model various times, sizes, measurements, and all other random variables that assume an entire interval of possible values.

Continuous distributions are described by their densities that play a role analogous to probability mass functions of discrete variables. Computing probabilities essentially reduces to integrating a density over the given set. Expectations and variances are defined similarly to the discrete case, replacing a probability mass function by a density and summation by integration.

In different situations, one uses Uniform, Exponential, Gamma, or Normal distributions. A few other families are studied in later chapters.

The Central Limit Theorem states that a standardized sum of a large number of independent random variables is approximately Normal, thus Table A4 can be used to compute related probabilities. A continuity correction should be used when a discrete distribution is approximated by a continuous distribution.

Characteristics of continuous families are summarized in Section 12.1.2.

Exercises

- 4.1.** The lifetime, in years, of some electronic component is a continuous random variable with the density

$$f(x) = \begin{cases} \frac{k}{x^4} & \text{for } x \geq 1 \\ 0 & \text{for } x < 1. \end{cases}$$

Find k , the cumulative distribution function, and the probability for the lifetime to exceed 2 years.

- 4.2.** The time, in minutes, it takes to reboot a certain system is a continuous variable with the density

$$f(x) = \begin{cases} C(10 - x)^2, & \text{if } 0 < x < 10 \\ 0, & \text{otherwise} \end{cases}$$

(a) Compute C .

(b) Compute the probability that it takes between 1 and 2 minutes to reboot.

- 4.3.** The installation time, in hours, for a certain software module has a probability density function $f(x) = k(1 - x^3)$ for $0 < x < 1$. Find k and compute the probability that it takes less than $1/2$ hour to install this module.

- 4.4.** Lifetime of a certain hardware is a continuous random variable with density

$$f(x) = \begin{cases} K - x/50 & \text{for } 0 < x < 10 \text{ years} \\ 0 & \text{for all other } x \end{cases}$$

(a) Find K .

(b) What is the probability of a failure within the first 5 years?

(c) What is the expectation of the lifetime?

- 4.5.** Two continuous random variables X and Y have the joint density

$$f(x, y) = C(x^2 + y), \quad -1 \leq x \leq 1, \quad 0 \leq y \leq 1.$$

(a) Compute the constant C .

(b) Find the marginal densities of X and Y . Are these two variables independent?

(c) Compute probabilities $P\{Y < 0.6\}$ and $P\{Y < 0.6 \mid X < 0.5\}$.

- 4.6.** A program is divided into 3 blocks that are being compiled on 3 parallel computers. Each block takes an Exponential amount of time, 5 minutes on the average, independently of other blocks. The program is completed when *all* the blocks are compiled. Compute the expected time it takes the program to be compiled.
- 4.7.** The time it takes a printer to print a job is an Exponential random variable with the expectation of 12 seconds. You send a job to the printer at 10:00 am, and it appears to be third in line. What is the probability that your job will be ready before 10:01?
- 4.8.** For some electronic component, the time until failure has Gamma distribution with parameters $\alpha = 2$ and $\lambda = 2$ (years⁻¹). Compute the probability that the component fails within the first 6 months.
- 4.9.** On the average, a computer experiences breakdowns every 5 months. The time until the first breakdown and the times between any two consecutive breakdowns are independent Exponential random variables. After the third breakdown, a computer requires a special maintenance.
- (a) Compute the probability that a special maintenance is required within the next 9 months.
- (b) Given that a special maintenance was not required during the first 12 months, what is the probability that it will not be required within the next 4 months?
- 4.10.** Two computer specialists are completing work orders. The first specialist receives 60% of all orders. Each order takes her Exponential amount of time with parameter $\lambda_1 = 3$ hrs⁻¹. The second specialist receives the remaining 40% of orders. Each order takes him Exponential amount of time with parameter $\lambda_2 = 2$ hrs⁻¹.
- A certain order was submitted 30 minutes ago, and it is still not ready. What is the probability that the first specialist is working on it?
- 4.11.** Consider a satellite whose work is based on a certain block A. This block has an independent backup B. The satellite performs its task until both A and B fail. The lifetimes of A and B are exponentially distributed with the mean lifetime of 10 years.
- (a) What is the probability that the satellite will work for more than 10 years?
- (b) Compute the expected lifetime of the satellite.
- 4.12.** A computer processes tasks in the order they are received. Each task takes an Exponential amount of time with the average of 2 minutes. Compute the probability that a package of 5 tasks is processed in less than 8 minutes.
- 4.13.** On the average, it takes 25 seconds to download a file from the internet. If it takes an Exponential amount of time to download one file, then what is the probability that it will take more than 70 seconds to download 3 independent files?
- 4.14.** The time X it takes to reboot a certain system has Gamma distribution with $\mathbf{E}(X) = 20$ min and $\text{Std}(X) = 10$ min.

- (a) Compute parameters of this distribution.
- (b) What is the probability that it takes less than 15 minutes to reboot this system?

4.15. A certain system is based on two independent modules, A and B. A failure of any module causes a failure of the whole system. The lifetime of each module has a Gamma distribution, with parameters α and λ given in the table,

Component	α	λ (years ⁻¹)
<i>A</i>	3	1
<i>B</i>	2	2

- (a) What is the probability that the system works at least 2 years without a failure?
- (b) Given that the system failed during the first 2 years, what is the probability that it failed due to the failure of component B (but not component A)?

4.16. Let Z be a Standard Normal random variable. Compute

- (a) $P(Z < 1.25)$
- (b) $P(Z \leq 1.25)$
- (c) $P(Z > 1.25)$
- (d) $P(|Z| \leq 1.25)$
- (e) $P(Z < 6.0)$
- (f) $P(Z > 6.0)$
- (g) With probability 0.8, variable Z does not exceed what value?

4.17. For a Standard Normal random variable Z , compute

- (a) $P(Z \geq 0.99)$
- (b) $P(Z \leq -0.99)$
- (c) $P(Z < 0.99)$
- (d) $P(|Z| > 0.99)$
- (e) $P(Z < 10.0)$
- (f) $P(Z > 10.0)$
- (g) With probability 0.9, variable Z is less than what?

4.18. For a Normal random variable X with $\mathbf{E}(X) = -3$ and $\text{Var}(X) = 4$, compute

- (a) $P(X \leq 2.39)$
- (b) $P(Z \geq -2.39)$
- (c) $P(|X| \geq 2.39)$
- (d) $P(|X + 3| \geq 2.39)$
- (e) $P(X < 5)$
- (f) $P(|X| < 5)$
- (g) With probability 0.33, variable X exceeds what value?

4.19. According to one of the Western Electric rules for quality control, a produced item is considered conforming if its measurement falls within three standard deviations from the target value. Suppose that the process is in control so that the expected value of each measurement equals the target value. What percent of items will be considered conforming, if the distribution of measurements is

- (a) $\text{Normal}(\mu, \sigma)$?
- (b) $\text{Uniform}(a, b)$?

4.20. Refer to Exercise 4.19. What percent of items falls *beyond* 1.5 standard deviations from

the mean, if the distribution of measurements is

- (a) $\text{Normal}(\mu, \sigma)$?
- (b) $\text{Uniform}(a, b)$?

4.21. The average height of professional basketball players is around 6 feet 7 inches, and the standard deviation is 3.89 inches. Assuming Normal distribution of heights within this group,

- (a) What percent of professional basketball players are taller than 7 feet?
- (b) If your favorite player is within the tallest 20% of all players, what can his height be?

4.22. Refer to the country in Example 4.11 on p. 91, where household incomes follow Normal distribution with $\mu = 900$ coins and $\sigma = 200$ coins.

- (a) A recent economic reform made households with the income below 640 coins qualify for a free bottle of milk at every breakfast. What portion of the population qualifies for a free bottle of milk?
- (b) Moreover, households with an income within the lowest 5% of the population are entitled to a free sandwich. What income qualifies a household to receive free sandwiches?

4.23. The lifetime of a certain electronic component is a random variable with the expectation of 5000 hours and a standard deviation of 100 hours. What is the probability that the average lifetime of 400 components is less than 5012 hours?

4.24. Installation of some software package requires downloading 82 files. On the average, it takes 15 sec to download one file, with a variance of 16 sec². What is the probability that the software is installed in less than 20 minutes?

4.25. Among all the computer chips produced by a certain factory, 6 percent are defective. A sample of 400 chips is selected for inspection.

- (a) What is the probability that this sample contains between 20 and 25 defective chips (including 20 and 25)?
- (b) Suppose that each of 40 inspectors collects a sample of 400 chips. What is the probability that at least 8 inspectors will find between 20 and 25 defective chips in their samples?

4.26. An average scanned image occupies 0.6 megabytes of memory with a standard deviation of 0.4 megabytes. If you plan to publish 80 images on your web site, what is the probability that their total size is between 47 megabytes and 50 megabytes?

4.27. A certain computer virus can damage any file with probability 35%, independently of other files. Suppose this virus enters a folder containing 2400 files. Compute the probability that between 800 and 850 files get damaged (including 800 and 850).

- 4.28.** Seventy independent messages are sent from an electronic transmission center. Messages are processed sequentially, one after another. Transmission time of each message is Exponential with parameter $\lambda = 5 \text{ min}^{-1}$. Find the probability that all 70 messages are transmitted in less than 12 minutes. Use the Central Limit Theorem.
- 4.29.** A computer lab has two printers. Printer I handles 40% of all the jobs. Its printing time is Exponential with the mean of 2 minutes. Printer II handles the remaining 60% of jobs. Its printing time is Uniform between 0 minutes and 5 minutes. A job was printed in less than 1 minute. What is the probability that it was printed by Printer I?
- 4.30.** An internet service provider has two connection lines for its customers. Eighty percent of customers are connected through Line I, and twenty percent are connected through Line II. Line I has a Gamma connection time with parameters $\alpha = 3$ and $\lambda = 2 \text{ min}^{-1}$. Line II has a Uniform(a, b) connection time with parameters $a = 20 \text{ sec}$ and $b = 50 \text{ sec}$. Compute the probability that it takes a randomly selected customer more than 30 seconds to connect to the internet.
- 4.31.** Upgrading a certain software package requires installation of 68 new files. Files are installed consecutively. The installation time is random, but on the average, it takes 15 sec to install one file, with a variance of 11 sec^2 .
- (a) What is the probability that the whole package is upgraded in less than 12 minutes?
 - (b) A new version of the package is released. It requires only N new files to be installed, and it is promised that 95% of the time upgrading takes less than 10 minutes. Given this information, compute N .
- 4.32.** Two independent customers are scheduled to arrive in the afternoon. Their arrival times are uniformly distributed between 2 pm and 8 pm. Compute
- (a) the expected time of the first (earlier) arrival;
 - (b) the expected time of the last (later) arrival.
- 4.33.** Let X and Y be independent Standard Uniform random variables.
- (a) Find the probability of an event $\{0.5 < (X + Y) < 1.5\}$.
 - (b) Find the conditional probability $P\{0.3 < X < 0.7 \mid Y > 0.5\}$.

This problem can be solved analytically as well as geometrically.

- 4.34.** Prove the memoryless property of Geometric distribution. That is, if X has Geometric distribution with parameter p , show that

$$P\{X > x + y \mid X > y\} = P\{X > x\}$$

for any integer $x, y \geq 0$.